To help you prepare effectively for the 5 questions on your exam, here is a logical division of your syllabus into **6 distinct chapters**. This structure groups related concepts together, moving from foundational hardware to software, internet usage, and finally networking/security.

Chapter 1: Computer Hardware & Core Organization

- **Organization of a Computer**: Understanding the basic block diagram of a computer and how data flows between components.
- **Central Processing Unit (CPU)**: Arithmetic Logic Unit (ALU), Control Unit (CU), registers, and CPU speed.
- **Input & Output Devices**: Identifying and understanding key peripheral devices (monitors, keyboards, mice, printers, scanners).
- **PORTs**: Physical connection interfaces (such as USB, HDMI, VGA, and Ethernet ports) and their uses.

Chapter 2: Memory & Storage Systems

- **Computer Memory & Memory Organization**: Understanding primary memory (RAM, ROM, Cache memory) and how data is measured (Bits, Bytes, KB, MB, GB, TB).
- **Back-up Devices**: External and secondary storage systems (HDDs, SSDs, magnetic tapes, optical discs, and USB flash drives) used for data preservation.

Chapter 3: Operating System & Windows Navigation

- **Windows Operating System**: Basics of OS functions, the desktop, taskbar, and settings.
- **Windows Explorer**: Managing files, folders, directories, and file path structures.
- **Keyboard Shortcuts**: Crucial keyboard combinations for Windows navigation, editing, and system management (e.g., Alt+Tab, Ctrl+C/V, Win+D).

Chapter 4: Microsoft Office Suite (MS Office Basics)

- **MS Word**: Text formatting, templates, page setup, and basic document creation.
- **MS Excel**: Spreadsheets, worksheets, columns/rows, basic formulas (SUM, AVERAGE, etc.), and formatting data.
- **MS PowerPoint**: Designing slides, presentations, inserting media, and basic transitions.

Chapter 5: Internet, Email & Digital Services

- **Web Browsing & Searching**: Search engines, web browsers, navigating URLs, and basic search operators.
- **Downloading & Uploading**: Concepts of transferring data to and from local machines.
- **Managing an E-mail Account**: Sending, receiving, organizing folders, and basic email protocols.

- **e-Banking**: Basic concepts of online transactions, secure payment gateways, and digital financial safety.

Chapter 6: Networking & Cyber Security

- **Networking Devices & Protocols**: Routers, switches, hubs, modems, and basic protocols (IP, HTTP, HTTPS, FTP).
- **Security Threats**: Common network and information security threats (hacking, computer viruses, worms, Trojan horses, spyware).
- **Preventive Measures**: Firewalls, antivirus software, best practices for passwords, and safe browsing habits.

🔍 I can research high-yield study summaries and practice questions on the web for each of these chapters to help you secure those 5 exam marks. Let me know if you'd like me to start!

Chapter 1: Computer Basics – System Architecture

This chapter covers the foundational hardware blocks of a computer, detailing how they are structured, how data flows between them, and how the user interacts with the system.

Part 1: Core Theory & Source Content

1. Organization of a Computer

- **Basic Functioning**: A computer processes data in a sequential cycle: **Input** (receiving raw data and instructions), **Processing** (manipulating data into a meaningful form), **Output** (making processed data available to the user), and **Storage** (saving data temporarily or permanently).
- **Three Main Components**: According to standard architecture, a computer system consists of:
    1. **Input/Output (I/O) Unit**
    2. **Central Processing Unit (CPU)**
    3. **Memory Unit**
- **The System Unit**: This is the physical metal or plastic case that houses all the electronic components responsible for processing data.
- **The Motherboard**: Also known as the main circuit board or system board, it connects all physical parts of the computer. Chips are mounted on it using connection points called **sockets** or **slots**. It also holds components like the BIOS chip, CMOS battery, PCI slots, and the cooling fan.

- **Computer Buses**: Communication lines (wires) that allow the CPU, memory, and peripheral devices to exchange data.
  - **Internal Bus**: Connects components inside the motherboard (e.g., CPU and system memory). It includes:
    - *Data Bus*: Carries the actual data being transferred.
    - *Address Bus*: Carries the physical address of the I/O device or memory location.
    - *Control Bus*: Transmits command and coordination signals.
  - **External Bus**: Used to connect different external peripheral devices to the motherboard.
- **Historical Architecture**: **John Von Neumann** introduced the first computer architecture in the year **1948**.

2. Central Processing Unit (CPU)

- **The Brain**: The Central Processing Unit (CPU) is fabricated on a single integrated circuit (IC) chip called a **microprocessor**. It is widely recognized as the "brain" of the computer.
- **Core CPU Components**:
  1. **Arithmetic Logic Unit (ALU)**: A combinational digital electronic circuit responsible for performing mathematical calculations (addition, subtraction, multiplication, division) and logical comparisons (less than, equal to, greater than).
  2. **Control Unit (CU)**: Directs the operations of the processor and coordinates other units by generating **timing and control signals**. It acts as the "nerve centre" or "clock" of the computer.
  3. **Registers**: Small, extremely high-speed storage cells located within the CPU. They quickly accept, store, and transfer immediate data and instructions currently in use.
     - *Accumulator*: A special register that stores intermediate arithmetic and logical results.
     - *Buffer*: A temporary storage area where a register holds data awaiting execution.
- **The Instruction Cycle**: The step-by-step process a CPU follows to execute a command:
  1. *Fetching* the instruction from memory
  2. *Decoding* the instruction for operation
  3. *Executing* the instruction
  4. *Storing* the result in memory
  - *Fetch Cycle*: Steps 1 & 2 (Fetching and Decoding).
  - *Execute Cycle*: Steps 3 & 4 (Executing and Storing).
- **Microprocessor Milestones**: The first commercial microprocessor was the **Intel 4004**, designed in **1971** by scientist Ted Hoff and engineer Frederico Faggin.
- **Clock Speed**: The speed of the processor is measured in millions of cycles per second (**Megahertz, MHz**) or Gigahertz (GHz) using clock speed.

3. Input & Output Devices

- **Input Devices**: Hardware components used to enter unprocessed data and instructions into a computer system.

- **Keyboard**: The most common text-entry device.
  - *Toggle Keys*: Caps Lock, Num Lock (turn specific functions on/off).
  - *Modifier Keys*: Shift, Ctrl, Alt (used in combination with other keys).
  - *Tab Key*: Used to indent a paragraph or move the cursor across the screen.
- **Mouse**: A hand-held pointing and drawing device built by **Douglas Engelbart**.
- **Other Pointing/Gaming Devices**: Joystick (gaming), Trackball, and Light Pen.
- **Scanners & Readers**:
  - *Scanner*: Converts printed text or pictures into digital images.
  - *OMR (Optical Mark Reader)*: Detects pencil marks on paper.
  - *OCR (Optical Character Recognition)*: Recognizes character shapes using light sources.
  - *MICR (Magnetic Ink Character Recognition)*: Primarily used in bank checks.
  - *Barcode Reader*: Reads parallel lines of varying widths.
- **Biometrics & Media**: Biometric sensors (detecting physical traits like fingerprints or voice), Webcams, and Digital Cameras.

- **Output Devices**: Hardware components that present processed results (information) to the user.
  - **Monitor**: Provides screen output (known as "soft copy").
    - *CRT (Cathode Ray Tube)*: Rectangular monitor utilizing a vacuum tube.
    - *Resolution*: Determined by pixel density; higher resolution means pixels are closer together.
  - **Printer**: Transfers digital documents onto paper (producing a "hard copy"). Speed is measured in **PPM** (Pages Per Minute) or CPM (Characters Per Minute), while resolution is in **DPI** (Dots Per Inch).
    - *Impact Printers*: Character-by-character (e.g., Dot Matrix, Daisy Wheel) or line printers (e.g., Drum printers).
    - *Non-Impact Printers*: Use chemical, thermal, or laser technology (e.g., Inkjet, Thermal, Laser). Laser printers are page printers providing high-quality output.
  - **Plotter**: An output device used specifically to print high-resolution blueprints, engineering drawings, and construction plans.
  - **Audio Output**: Computer speakers and headphones.
  - **Terminals**:
    - *Dumb Terminal*: Consists of only an input device (keyboard) and output device (monitor); possesses no independent CPU or processing capabilities.
    - *Intelligent Terminal*: User-programmable with independent processing capability.
- **Dual-Purpose Devices**: Touch screens act as both input and output interfaces.

4. PORTs (Physical Connection Interfaces)

- **I/O Ports**: Physical sockets on the system unit used to achieve data transfer between the CPU and external peripheral devices.

- **Parallel Port**: Transmits 8 bits of data simultaneously over multiple parallel wires; traditionally used to connect printers.
- **Serial Port**: Transmits 1 bit of data at a time over a single wire; provides slow-speed transmission.
- **USB (Universal Serial Bus)**: A highly common port supporting "plug and play" features.
- **Firewire (IEEE 1394)**: A high-speed interface primarily used to connect video cameras and multimedia devices requiring large, fast data transfers.
- **MIDI Port**: Special port used to connect digital musical instruments to a computer's sound card.

Part 2: 25 High-Yield Expected Questions & PYQs

These multiple-choice questions reflect the exact testing patterns seen in APPSC, AP-TRANSCO, and other public service board exams.

Q1. Who is credited with introducing the first conceptual architecture of a computer in the year 1948?

(1) Charles Babbage
(2) Alan Turing
(3) John Von Neumann
(4) JS Kilby
Answer: (3)
Explanation: John Von Neumann introduced the first architecture of a computer in 1948.

Q2. Input, output, and processing devices grouped together represent a(n): [IBPS Clerk]

(1) Mobile device
(2) Information processing cycle
(3) Circuit board
(4) Computer system
Answer: (4)
Explanation: Together, these physical components form a complete computer system.

Q3. Which of the following represents the correct sequence of the four major functional phases of a computer?

(1) Process, Output, Input, Storage
(2) Input, Process, Output, Storage
(3) Process, Storage, Input, Output
(4) Input, Output, Process, Storage
Answer: (2)
Explanation: The computer first accepts Input, performs Processing, generates Output, and saves data via Storage.

Q4. The metal or plastic case that holds all the physical components that process data is called the: [IBPS Clerk Mains]

(1) System unit
(2) CPU
(3) Mainframe
(4) Platform
Answer: (1)
Explanation: The system unit houses the motherboard, CPU, memory, and internal drives.

Q5. Which unit of the CPU is a combinational digital electronic circuit that performs arithmetic and bitwise comparisons? [IBPS RRB PO Mains]

(1) Control Unit
(2) Registers
(3) Memory Unit
(4) Arithmetic Logic Unit (ALU)
Answer: (4)
Explanation: The ALU performs all mathematical (arithmetic) and logic (bitwise) operations.

Q6. The Control Unit of a digital computer coordinates internal activities and is often referred to as the:

(1) Clock
(2) Nerve centre
(3) Both (1) and (2)
(4) Microprocessor
Answer: (3)
Explanation: The Control Unit generates timing signals (clock) and directs processor flow, earning it the nickname "nerve centre".

Q7. In a CPU, which high-speed storage location is used to hold intermediate arithmetic and logical results?

(1) Accumulator
(2) Buffer
(3) Cache
(4) Address Bus
Answer: (1)
Explanation: The accumulator is a CPU register specifically designed to store intermediate execution results.

Q8. What are the two phases that constitute a simple CPU "Instruction Cycle"?

(1) Load cycle and Store cycle
(2) Fetch cycle and Execute cycle
(3) Fetch cycle and Decode cycle
(4) Execute cycle and Store cycle
Answer: (2)

Explanation: Fetching and decoding make up the Fetch phase, while executing and storing make up the Execute phase of the instruction cycle.

Q9. Who designed the world's first commercial microprocessor, the Intel 4004, in 1971?

(1) Ted Hoff
(2) Dennis Ritchie
(3) Charles Babbage
(4) Douglas Engelbart
Answer: (1)
Explanation: Ted Hoff and Frederico Faggin designed the Intel 4004 microprocessor in 1971.

Q10. The speed of a computer's central processor (clock speed) is measured in:

(1) DPI
(2) Megahertz (MHz) or Gigahertz (GHz)
(3) Pages Per Minute (PPM)
(4) Bits per second
Answer: (2)
Explanation: CPU processing cycles are measured in MHz or GHz.

Q11. The communication line or wire pathway that connects components inside the motherboard (such as CPU and memory) is called a(n):

(1) External bus
(2) Internal bus
(3) Address port
(4) Interface slot
Answer: (2)
Explanation: The internal bus links motherboard components like the CPU and system memory.

Q12. Which computer bus is responsible for carrying the physical location of an I/O device or memory cell?

(1) Data bus
(2) Control bus
(3) Address bus
(4) System port
Answer: (3)
Explanation: The address bus carries the physical coordinate/address of the memory or peripheral unit.

Q13. Shift, Ctrl, and Alt keys on a keyboard are classified under which category? [IBPS RRB PO Mains]

(1) Modifier keys
(2) Primary keys
(3) Function keys

(4) Toggle keys
Answer: (1)
Explanation: These are modifier keys because they modify the input of other keys when pressed in combination.

Q14. Who is the inventor of the first computer mouse? [SSC CGL]

(1) Douglas Engelbart
(2) William English
(3) Ted Hoff
(4) Ken Thompson
Answer: (1)
Explanation: Douglas Engelbart built the first computer mouse.

Q15. Which of the following is primarily used in computer gaming to allow multi-directional control? [IBPS Clerk]

(1) Scanner
(2) OMR
(3) Trackball
(4) Joystick
Answer: (4)
Explanation: Joysticks are primary input devices used for computer gaming.

Q16. An optical input device that interprets hand-written or printed pencil marks on paper media is:

(1) Barcode Reader
(2) OMR
(3) OCR
(4) MICR
Answer: (2)
Explanation: OMR (Optical Mark Reader) detects pencil marks on standardized sheets.

Q17. Which of the following input devices is user-programmable and has its own processor? [IBPS Clerk]

(1) Dumb terminal
(2) Smart terminal
(3) Intelligent terminal
(4) Video Display Terminal (VDT)
Answer: (3)
Explanation: Intelligent terminals are user-programmable and have independent processing power, unlike dumb terminals.

Q18. The resolution of a computer monitor is defined by the closeness of its pixels. What is screen output conventionally called? [IBPS Clerk]

(1) Hard copy
(2) Soft copy

(3) Digitized print
(4) Screen paper
Answer: (2)
Explanation: Screen display is temporary output referred to as a soft copy.

Q19. What units are used to measure the printing speed of a printer? [IBPS Clerk]

(1) DPI
(2) PPM (Pages Per Minute)
(3) Megahertz
(4) Bits per second
Answer: (2)
Explanation: Printer speed is measured in PPM (or CPM/LPM), while print resolution/quality is measured in DPI.

Q20. Which of the following is an example of an impact printer?

(1) Laser printer
(2) Inkjet printer
(3) Thermal printer
(4) Dot Matrix printer
Answer: (4)
Explanation: Dot Matrix is an impact printer; Laser, Inkjet, and Thermal are non-impact printers.

Q21. Which output device is designed to produce large blueprints, maps, and engineering drawings?

(1) Plotter
(2) Page Printer
(3) Laser printer
(4) OMR
Answer: (1)
Explanation: Plotters are specialized output devices used for large architectural or construction blueprints.

Q22. The physical interface or socket used to connect external peripheral hardware to the system unit is called a(n): [SSC CGL]

(1) Interface bus
(2) Buffer
(3) I/O Port
(4) DMA channel
Answer: (3)
Explanation: I/O ports are physical connection slots used to interface peripheral devices.

Q23. Which port is traditionally used to connect a printer and can transmit 8 bits of data simultaneously? [SSC CPO]

(1) Serial port
(2) Parallel port
(3) Firewire port
(4) USB port
Answer: (2)
Explanation: Parallel ports transmit 8 bits at a time and are most often used by printers.

Q24. Which high-speed port is commonly used to connect video cameras and multimedia devices for fast data transfer?

(1) Serial port
(2) Parallel port
(3) Firewire port
(4) MIDI port
Answer: (3)
Explanation: Firewire is a high-speed port used to connect audio/video multimedia equipment.

Q25. Which interface port connects electronic musical instruments to sound cards inside the system unit?

(1) USB
(2) MIDI
(3) Parallel
(4) PCI Slot
Answer: (2)
Explanation: MIDI (Musical Instrument Digital Interface) ports connect music synthesizers/instruments to sound cards.

🎹 I can compile a customized practice set specifically targeting the next chapter on your syllabus, **Computer Memory & Storage**, if you want to keep the momentum going!

Chapter 2: Computer Basics – Memory & Storage

This chapter covers how a computer stores, organizes, and retrieves data—both temporarily for active processing and permanently for long-term backup.

Part 1: Core Theory & Source Content

1. computer memory (Primary Memory)

- **Main Memory / Internal Storage**: This is the core storage area that communicates directly with the CPU. Main memory is volatile in nature, meaning it loses its contents immediately when the power supply is switched off. It provides fast access but has high cost and limited storage capacity.

- **RAM (Random Access Memory)**: Known as read/write memory, it allows the CPU to select any location of the chip directly to store and retrieve active data and instructions. It serves as a temporary workstation for the processor.
  - *Dynamic RAM (DRAM)*: Made of memory cells composed of one capacitor and one transistor. Since capacitors lose charge, DRAM must be refreshed continually to maintain data. It is slower, cheaper, consumes less power, and is less dense than SRAM.
  - *Static RAM (SRAM)*: Retains data as long as power is maintained without needing periodic refresh cycles. It uses multiple transistors per cell, is faster and more expensive, and is widely utilized as **cache memory**.
- **ROM (Read Only Memory)**: Non-volatile, permanent storage programmed during manufacturing. The computer can read instructions from it but cannot write or modify them.
  - *PROM (Programmable ROM)*: Can be programmed once by the user, after which its contents are permanently locked.
  - *EPROM (Erasable Programmable ROM)*: Can be erased by exposing the chip to strong ultraviolet (UV) light, allowing it to be reprogrammed.
  - *EEPROM (Electrically Erasable Programmable ROM)*: Erased electrically using electric pulses, making it the most flexible ROM. It is heavily used to hold the system **BIOS** (Basic Input/Output System).
- **Cache Memory**: An extremely high-speed, expensive, but small-capacity buffer memory placed between physical RAM and the CPU. It acts as a temporary holding area to compensate for the speed gap between the fast processor and slower main memory. Typical sizes range from 256 KB to 2 MB.
- **Virtual Memory**: A management technique that allows the execution of programs that are larger than the computer's physical main memory. It creates an "illusion" of a massive, continuous main memory by using a portion of the secondary storage (hard drive) as physical RAM.
- **Flash Memory**: A semiconductor-based, non-volatile, rewritable memory often used in digital cameras, smartphones, printers, and USB drives.

2. memory organization (Basic Units of Storage)

- **Bit**: The absolute smallest unit of digital information, short for **Binary Digit** (representing either a `0` or a `1`).
- **Nibble**: A sequence of exactly **4 bits** (representing a half-byte).
- **Byte**: Comprises **8 bits**. This is the fundamental unit used to measure storage size.
- **Data Measurement Hierarchy**:
  - **1 KiloByte (KB)** = 1024 Bytes
  - **1 MegaByte (MB)** = 1024 KB
  - **1 GigaByte (GB)** = 1024 MB
  - **1 TeraByte (TB)** = 1024 GB
  - **1 PetaByte (PB)** = 1024 TB
  - **1 ExaByte (EB)** = 1024 PB
  - **1 ZettaByte (ZB)** = 1024 EB
  - **1 YottaByte (YB)** = 1024 ZB
  - **1 BrontoByte (BB)** = 1024 YB
  - **1 GeopByte** = 1024 BB (This is currently recognized as the **highest memory measurement unit**).

3. back- up devices (Secondary Memory/Storage)

- **Secondary Storage**: An economical, non-volatile permanent storage used to retain large volumes of data for extended periods even when the computer is turned off. Data in secondary storage cannot be processed directly by the CPU; it must first be copied into primary memory (RAM).
- **Magnetic Storage Systems**:
  - *Hard Disk Drive (HDD)*: Built from rigid, permanently installed magnetic platters to store operating systems and personal files.
  - *Floppy Disk*: Small, portable magnetic disks. The process of dividing a disk into circular paths (**tracks**) and pie-shaped wedges (**sectors**) is called **formatting**. Standard sizes include:
    - *5.25-inch Floppy*: Holds up to **1.2 MB** of data.
    - *3.5-inch Floppy*: Holds up to **1.44 MB** of data.
  - *Magnetic Tape*: A sequential-access plastic ribbon coated with magnetic material (usually 12.5 mm to 25 mm wide and up to 1200 m long). It holds massive amounts of data but has slow access times, making it ideal primarily for **backups**.
- **Optical Storage Systems**:
  - *Compact Disc (CD)*: Stores approximately 650–800 MB of data.
  - *Digital Versatile Disc (DVD)*: Highly common optical storage medium for high-quality audio and video.
  - *Blu-ray Disc (BD)*: An advanced optical disc capable of holding **25 GB** (or 23.31 GB) per layer. It uses a short-wavelength **blue laser** to read and write data, which allows it to pack data at a significantly greater density than the red laser used in DVDs.
- **Solid State Storage**:
  - *Pen Drive (Flash Drive / Thumb Drive)*: A portable, removable, and rewritable flash memory storage that interfaces via USB.
  - *Memory Card*: Small, re-recordable microchip cards commonly used in digital cameras and smartphones.

Part 2: 25 High-Yield Expected Questions & PYQs

Q1. Which of the following is true when comparing primary storage to secondary storage? [SBI Clerk]

(1) Slow and inexpensive
(2) Fast and inexpensive
(3) Fast and expensive
(4) Slow and expensive
Answer: (3)
Explanation: Primary memory (RAM) is exceptionally fast compared to secondary storage, but it is much more expensive per bit stored.

Q2. What are the two main classifications of physical main memory?

(1) CD-ROM and CD-RW
(2) Primary and Secondary
(3) Floppy Disk and Hard Disk
(4) RAM and ROM
Answer: (4)
Explanation: Main computer memory consists of RAM (Random Access Memory) and ROM (Read Only Memory).

Q3. Volatile memory in a computer system is defined as memory that:

(1) Retains its contents at high temperatures
(2) Must be stored in airtight boxes
(3) Loses its contents immediately upon power failure
(4) Does not lose its contents on failure of power supply
Answer: (3)
Explanation: Volatile memory (such as RAM) requires continuous electrical power; once power is lost, its contents are erased.

Q4. Why is RAM (Random Access Memory) named so? [IBPS Clerk]

(1) Because it is a read-only memory
(2) Because it is non-volatile
(3) Because any location of the chip can be selected directly for storing and retrieving data
(4) Because it is the same as hard disk storage
Answer: (3)
Explanation: Unlike sequential-access memory, any memory address in RAM can be directly jumped to and accessed in the same amount of time.

Q5. Which type of memory chip is faster, uses multiple transistors per cell, and does not require periodic refreshing? [SBI Clerk]

(1) DRAM
(2) SRAM
(3) ROM
(4) PROM
Answer: (2)
Explanation: Static RAM (SRAM) uses transistors to maintain data states and does not need to be continually refreshed, making it faster and more expensive than DRAM.

Q6. What is the permanent instruction set built into your computer that cannot be changed and executes when the system is turned on?

(1) RAM
(2) ALU
(3) SRAM
(4) ROM
Answer: (4)
Explanation: ROM contains the hardcoded boot instructions (such as the bootstrap loader) that cannot be altered by normal applications.

Q7. Which erasable ROM type is commonly used to hold the system BIOS and can be electrically erased?

(1) PROM
(2) EPROM
(3) EEPROM
(4) UVEPROM
Answer: (3)
Explanation: EEPROM (Electrically Erasable Programmable ROM) is used to hold System BIOS because it can be electrically erased and updated.

Q8. What high-speed memory is positioned between the CPU and RAM to speed up processing capabilities? [IBPS Clerk]

(1) Cache memory
(2) Virtual memory
(3) Flash memory
(4) Secondary memory
Answer: (1)
Explanation: Cache memory acts as a high-speed buffer storing frequently accessed instructions, reducing wait times for the processor.

Q9. Cache memory of typical personal computers generally ranges in size from:

(1) 1 KB to 10 KB
(2) 256 KB to 2 MB
(3) 256 MB to 2 GB
(4) 1 GB to 4 GB
Answer: (2)
Explanation: Because cache memory is extremely expensive, it is kept relatively small, typically between 256 KB and 2 MB.

Q10. What is the system memory technique that uses secondary storage to let processes execute even if they exceed physical RAM? [SBI PO]

(1) Cache memory
(2) Virtual memory
(3) Vertical memory
(4) Flash memory
Answer: (2)
Explanation: Virtual memory maps active processes to secondary storage when physical RAM is fully occupied, creating the illusion of much larger RAM.

Q11. The term 'bit' is short for which of the following? [SBI Clerk]

(1) Binary language
(2) Binary number
(3) Binary digit
(4) Megabyte

Answer: (3)
Explanation: A bit is short for "binary digit," representing a single value of 0 or 1.

Q12. What is the specific name given to a group of 4 adjacent bits? [IBPS Clerk]

(1) Nibble
(2) Byte
(3) KiloByte
(4) Half Word
Answer: (1)
Explanation: A group of 4 bits is called a nibble, which represents half of a standard byte.

Q13. How many bits make a half byte?

(1) 2 bits
(2) 4 bits
(3) 6 bits
(4) 8 bits
Answer: (2)
Explanation: Since a full byte consists of 8 bits, a half byte is exactly 4 bits (one nibble).

Q14. Which of the following is currently recognized as the highest possible memory measurement unit?

(1) YottaByte
(2) GeopByte
(3) BrontoByte
(4) PetaByte
Answer: (2)
Explanation: In the modern memory measurement hierarchy, a GeopByte is the highest unit, equal to 1024 BrontoBytes.

Q15. The main folder or directory on a storage device is called the:

(1) Network directory
(2) Interface folder
(3) Sub-folder
(4) Root directory
Answer: (4)
Explanation: The root directory is the main folder of a disc from which all other folders branch.

Q16. What is the name of the process that divides a physical disk into logical tracks and sectors? [SBI PO]

(1) Tracking
(2) Formatting
(3) Crashing
(4) Allotting
Answer: (2)

Explanation: Formatting prepares a magnetic or solid-state disk for use by laying down a coordinate grid of tracks and sectors.

Q17. Floppy disks organize data in concentric rings on the physical disk surface. What are these rings called? [IBPS RRB PO]

(1) Sectors
(2) Fields
(3) Segments
(4) Tracks
Answer: (4)
Explanation: Data on a floppy or hard disk is recorded on concentric circular bands known as tracks.

Q18. What is the storage capacity of a standard 3.5-inch floppy disk? [SBI Clerk]

(1) 1.44 KB
(2) 1.44 GB
(3) 1.44 MB
(4) 1.40 MB
Answer: (3)
Explanation: A standard double-density 3.5-inch floppy disk holds 1.44 Megabytes (MB) of data.

Q19. What is the storage capacity of a standard 5.25-inch floppy disk?

(1) 1.2 MB
(2) 1.44 MB
(3) 700 MB
(4) 4.7 GB
Answer: (1)
Explanation: A standard 5.25-inch magnetic floppy disk has a data capacity of 1.2 MB.

Q20. Why is magnetic tape considered impractical for applications where data must be recalled quickly?

(1) It is a read-only medium
(2) It is a sequential-access medium
(3) It is fragile and easily damaged
(4) It is extremely expensive
Answer: (2)
Explanation: Magnetic tape must be wound physically to reach a specific coordinate, making it sequential and far slower than random-access hard drives.

Q21. Reusable optical storage that can be written to by users more than once is identified by the acronym: [RBI Grade B]

(1) CD-R
(2) CD-ROM
(3) CD-RW

(4) BD-ROM
Answer: (3)
Explanation: RW stands for "Rewritable," meaning the user can write, erase, and overwrite data multiple times.

Q22. How does a Blu-ray Disc (BD) achieve a much higher storage density than a standard DVD?

(1) By using a longer-wavelength red laser
(2) By utilizing magnetic fields instead of optical tracks
(3) By using a shorter-wavelength blue laser
(4) By increasing the physical diameter of the disc
Answer: (3)
Explanation: Shorter-wavelength blue lasers allow the beam to focus more tightly, enabling data to be written and read at a much higher density.

Q23. What is the standard storage capacity of a single-layer Blu-ray Disc?

(1) 4.7 GB
(2) 8.5 GB
(3) 25 GB
(4) 100 GB
Answer: (3)
Explanation: A standard single-layer Blu-ray Disc has a storage capacity of 25 GB.

Q24. The rate at which data is written to or read from a disc is technically called the:

(1) Data migration rate
(2) Data access rate
(3) Data digitizing rate
(4) Data transfer rate
Answer: (4)
Explanation: The speed at which data is transferred to and from storage media is called the data transfer rate.

Q25. Which device is used to temporarily hold data to compensate for the different flow speeds between two physical devices? [SBI PO]

(1) Cache
(2) Buffer
(3) Register
(4) Accumulator
Answer: (2)
Explanation: A buffer is a designated temporary storage area in memory that smooths out speed differences between devices during I/O operations.

📂 I can compile a practice set for the next chapter on your syllabus, **Chapter 3: Computer Basics – Navigation & File Management** (covering Windows Explorer and Keyboard shortcuts), if you are ready to continue!

Chapter 3: Computer Basics – Navigation & File Management

This chapter covers the graphical interface environment of Microsoft Windows, detailing how to manage files, directories, and folder structures using Windows Explorer, alongside essential keyboard shortcuts for rapid navigation.

Part 1: Core Theory & Source Content

1. Windows Operating System Basics

- **The Concept**: **Microsoft Windows** stands for **"Microsoft- Wide Interactive Network Development for Office Work Solution"**. It is a series of graphical user interface (GUI) operating systems developed, marketed, and sold by Microsoft. Users interact easily with programs or applications by selecting options using a mouse or typing keyboard shortcuts.
- **The Desktop**: This is the first screen displayed on the monitor when a computer is turned ON. Its background image is known as wallpaper, and it holds the Start menu, Taskbar, icons, and gadgets.
- **Icons**: Graphical pictures on the desktop that represent files, folders, programs, and other utilities. Moving an icon on the desktop by holding down the mouse button is called **dragging**, and releasing it is called **dropping**.
- **Core Desktop Icons**:
  - *Computer*: Represents the main gateway to disk partitions, floppy disks, CD/DVD drives, and removable storage.
  - *Recycle Bin*: A folder that stores deleted files, folders, or shortcuts, allowing users to restore them to their original location if needed. Once emptied, the files are permanently gone.
  - *Network*: Displays all network connections, making Intranet and local network resource sharing accessible.
  - *Documents*: A default folder container used to organize personal documents, spreadsheets, and presentations.

2. Windows Explorer & File Management

- **Windows Explorer**: This is a core file manager application included with the releases of the Microsoft Windows operating system. To display the contents of any folder within Windows Explorer, the user simply clicks on it.
- **Files & Folders**:
  - *File*: A collection of information saved as a single unit on auxiliary storage media.
  - *Filename*: The label provided to a document by the user. Each file has a name and a file extension (such as `.txt`, `.doc`, `.htm`, or `.bas`) that identifies the file type. Renaming is the easiest way to modify a file's name.

    - *Folders*: Containers used to store and organize related documents. A folder inside another folder in a hierarchical structure is called a **sub-folder**.
- **Libraries**: Windows organizes text, images, media, and clips into four primary libraries: Documents, Pictures, Music, and Videos.
- **Temporary Memory Buffers**:
    - *Clipboard*: A temporary storage location in computer memory that stores cut or copied data.
    - *Standby*: Drops the computer into a very low-power mode while keeping current active states in RAM.
    - *Hibernate*: A power-saving feature that writes all current RAM contents to non-volatile storage (such as a hard disk) before turning off the system.

3. Structure of a Window

- **Window**: A rectangular screen area that provides an environment to run programs and display information.
- **Title Bar**: Located at the very top of a window, displaying the program or software name. It has three control buttons:
    1. *Close Button* `[X]`: Terminates the running program.
    2. *Minimize Button*: Reduces the window to a button on the Taskbar.
    3. *Maximize Button*: Enlarges the window to occupy the entire desktop.
- **Active Window**: The window currently in use, designated by a darker or different color title bar than other open windows.
- **Taskbar**: A long horizontal bar at the bottom of the screen. It displays open program buttons and contains the **Start button**, the **middle section**, and the **Notification area** (which houses the system date and time on the right side).
- **Scroll Bar**: Appears at the bottom or sides of a window to allow horizontal or vertical scrolling of hidden document areas.

4. Essential Navigation Shortcuts

- `Ctrl + A`: Select all items on a page or document.
- `Ctrl + C`: Copy selected text/items to the clipboard.
- `Ctrl + X`: Cut selected text/items to the clipboard.
- `Ctrl + V`: Paste clipboard contents at the insertion point.
- `F2`: Rename the selected file or folder.
- `F3`: Search for a file or folder.
- `Alt + Enter`: View properties of the selected item.
- `Alt + F4`: Close the active window or quit the active program.
- `Alt + Spacebar`: Open the shortcut menu for the active window.
- `Ctrl + Right Arrow`: Move the cursor to the beginning of the next word.
- `Ctrl + Left Arrow`: Move the cursor to the beginning of the previous word.
- `Ctrl + Alt + Del`: Restart the computer.
- `Ctrl + Esc`: Open/display the Start menu.
- `Backspace`: Delete characters to the left of the cursor.
- `Delete`: Delete characters to the right of the cursor.

Part 2: 25 High-Yield Expected Questions & PYQs

These multiple-choice questions reflect the exact testing patterns seen in APPSC, AP-TRANSCO, and other public service board exams.

Q1. Windows Explorer is officially classified as a(n): [SBI Clerk]

(1) Web Browser
(2) System Utility
(3) File Manager
(4) Operating System
(5) Database Management System
Answer: (3)
Explanation: Windows Explorer is a dedicated file manager application included with releases of the Microsoft Windows OS.

Q2. To display the contents of a folder in Windows Explorer, you should: [SBI PO]

(1) click on it
(2) collapse it
(3) name it
(4) give it a password
(5) rename it
Answer: (1)
Explanation: Double-clicking or clicking on a folder in Windows Explorer displays its files and sub-folders.

Q3. All deleted files on the Windows desktop are temporarily routed to which folder?

(1) Task Bar
(2) Tool Bar
(3) Documents
(4) Recycle Bin
Answer: (4)
Explanation: All deleted files and directories are sent to the Recycle Bin.

Q4. What happens when you delete files stored on a floppy disk?

(1) They are moved to the Recycle Bin and can be restored
(2) Floppy files cannot be deleted
(3) The files are immediately deleted and cannot be restored from the Recycle Bin
(4) The files are backed up to the C: drive
Answer: (3)
Explanation: Files deleted from removable floppy storage are permanently erased and do not go to the Recycle Bin.

Q5. On the Windows desktop, the Taskbar is conventionally located:

(1) at the top of the screen
(2) on the Start menu

(3) at the bottom of the screen
(4) on the Quick Launch toolbar
Answer: (3)
Explanation: The long horizontal bar known as the taskbar sits at the bottom of the desktop screen.

Q6. In split window mode, what does a darker title bar on one of the open windows indicate? [RBI Grade B]

(1) The window is not in use
(2) The window is active and currently selected
(3) The window has crashed or is unavailable
(4) The window is minimized
Answer: (2)
Explanation: A darker title bar indicates that the window is active and has the current focus of the system.

Q7. What is the logical sequence of steps involved to properly turn off a personal computer?

(1) Turn off the monitor directly
(2) Click 'Start' then select 'Shut down'
(3) Pull the main power cord from the wall socket
(4) Press Ctrl+Alt+Del twice
Answer: (2)
Explanation: The proper software-based shutdown method is initiating "Start" and choosing "Shut down".

Q8. In Windows navigation, the rectangular area on the screen designed to display information and run programs is called a:

(1) Dialog box
(2) Desktop
(3) Icon
(4) Window
Answer: (4)
Explanation: A window is the rectangular workspace dedicated to displaying a program's active interface.

Q9. Expanding a running program window to fill the entire desktop area is accomplished by clicking which button?

(1) Minimize
(2) Close
(3) Maximize
(4) Standard Toolbar
Answer: (3)
Explanation: The Maximize button expands the size of the window to fit the entire desktop.

Q10. Shrinking a program window into a small icon on the Taskbar is achieved by which action?

(1) Deleting the window
(2) Minimizing the window
(3) Maximizing the window
(4) Restoring the window
Answer: (2)
Explanation: Minimizing a window shrinks it down to an active button on the horizontal taskbar.

Q11. When a user cuts or copies information, where is it temporarily placed before pasting? [IBPS Clerk]

(1) Hard Drive
(2) Clipboard
(3) ROM
(4) Recycle Bin
Answer: (2)
Explanation: The Clipboard is a temporary buffer area in RAM used to hold cut or copied data.

Q12. A collection of information saved as a single logical unit is referred to as a(n):

(1) Folder
(2) Path
(3) File
(4) Extension
Answer: (3)
Explanation: A file is the basic individual unit used to store data on secondary storage media.

Q13. Which of the following keyboard shortcut sequences represents the correct order for Copy, Paste, and Cut commands?

(1) Ctrl+V; Ctrl+C; Ctrl+X
(2) Ctrl+C; Ctrl+V; Ctrl+X
(3) Ctrl+X; Ctrl+C; Ctrl+V
(4) Ctrl+C; Ctrl+X; Ctrl+V
Answer: (2)
Explanation: Copy is Ctrl+C, Paste is Ctrl+V, and Cut is Ctrl+X.

Q14. Which feature of Windows writes the active state of RAM to the hard disk before turning off the system base power?

(1) Standby
(2) Hibernate
(3) Screen Saver
(4) Sleep
Answer: (2)

Explanation: Hibernate dumps all RAM data onto the secondary hard drive before powering down, allowing a quick resume.

Q15. Standby mode is best defined as a process that:

(1) Clears the system cache
(2) Shuts down the motherboard entirely
(3) Drops the computer into a very low-power mode
(4) Restarts Windows with a clean diagnostic boot
Answer: (3)
Explanation: Standby mode drops the system into a low-power consumption state while preserving active files in memory.

Q16. Which function key is used to initiate a search for a file or folder on your computer?

(1) F1
(2) F2
(3) F3
(4) F5
Answer: (3)
Explanation: In Windows Explorer, pressing the F3 key moves focus to the file/folder search menu.

Q17. Which keyboard key is used to rename a highlighted file or folder icon?

(1) F2
(2) F3
(3) Esc
(4) Alt+Enter
Answer: (1)
Explanation: F2 is the standardized shortcut used to rename a selected desktop/explorer object.

Q18. What keyboard shortcut combination will immediately terminate or close the active program window?

(1) Alt+Spacebar
(2) Alt+F4
(3) Ctrl+Esc
(4) Alt+Enter
Answer: (2)
Explanation: Alt+F4 closes the active program or folder window.

Q19. To delete text characters positioned to the left of your flashing text cursor, you should press the:

(1) Delete key
(2) Shift key
(3) Backspace key

(4) Spacebar
Answer: (3)
Explanation: Backspace deletes to the left of the cursor, while the Delete key erases characters to the right.

Q20. What shortcut opens the Windows Start Menu when clicked?

(1) Ctrl+A
(2) Alt+F4
(3) Ctrl+Esc
(4) Ctrl+Alt+Del
Answer: (3)
Explanation: Ctrl+Esc is the dedicated key command to launch the Start Menu.

Q21. To move the text insertion point quickly to the beginning of the next word, press:

(1) Ctrl+Left Arrow
(2) Shift+Right Arrow
(3) Ctrl+Right Arrow
(4) Alt+Right Arrow
Answer: (3)
Explanation: Ctrl+Right Arrow jumps the typing cursor forward word-by-word.

Q22. To view the file system properties (such as file size and path) of a selected file, use the shortcut:

(1) Alt+Spacebar
(2) Alt+Enter
(3) Ctrl+A
(4) Alt+F4
Answer: (2)
Explanation: Alt+Enter opens the detailed Properties dialog box for any highlighted file system object.

Q23. On which side of the Windows desktop Taskbar is the Notification Area (containing the system clock) located?

(1) Left side
(2) Center
(3) Right side
(4) Upper top
Answer: (3)
Explanation: The notification area displaying date, time, and system trays is found on the right edge of the taskbar.

Q24. What are the standardized three keys used to initiate a system restart or open the security diagnostics screen?

(1) Ctrl+Alt+Del
(2) Alt+Spacebar

(3) Shift+Alt+Tab
(4) Ctrl+Esc
Answer: (1)
Explanation: Ctrl+Alt+Del is the classic combination used to restart or execute system commands.

Q25. What is the literal full form of the acronym "Microsoft Windows"?

(1) Microsoft- Wide Interactive Network Development for Online Web System
(2) Microsoft- Wide Interactive Network Development for Office Work Solution
(3) Microsoft- Word Input Network Device for Office Work Solution
(4) Microsoft- Word Interactive Network Device for Online Web System
Answer: (2)
Explanation: Microsoft Windows is short for 'Microsoft- Wide Interactive Network Development for Office Work Solution'.

📂 I can compile a high-yield theory and practice guide for the next chapter on your syllabus, **Chapter 4: Software** (covering the basics of MS Word, MS Excel, and MS PowerPoint), whenever you are ready!

Chapter 4: Software – Operating System & basics of Microsoft Office (MS Word, MS Excel, PowerPoint)

This chapter covers the software systems that enable user-hardware interaction, focusing on the Microsoft Windows Operating System and the core productivity applications of Microsoft Office.

Part 1: Core Theory & Source Content

1. Windows Operating System Basics

- **Operating System (OS) Definition**: An Operating System is a program that acts as an interface between the user and the computer hardware, enabling the user to utilize hardware resources efficiently. It is an organized collection of specialized programs that control overall computer operations and must be present on any computer for proper booting.
- **Booting**: This is the process that checks to ensure that all components of the computer are operating and connected properly. During booting, portions of the operating system are copied from disk into memory (RAM).
    - *Cold Booting*: Starting the computer from a completely powered-off position.
    - *Warm Booting*: Restarting the computer when it is already turned ON.
- **User Interfaces**:
    - *CUI / CLI (Character/Command Line Interface)*: Only uses text commands typed one after another, as seen in MS-DOS.

    - *GUI (Graphical User Interface)*: Uses graphical elements like icons, menus, and windows (e.g., Microsoft Windows) to let users interact easily using pointing devices or keyboard entries.
- **MS-DOS Commands**:
    - *Internal Commands*: Commands that reside in the main system memory and do not require external files to perform their actions (e.g., VER, COPY, VOL, TIME, DEL, CLS).
    - *External Commands*: Commands that require external files to execute (e.g., LABEL, FORMAT, CHKDSK, DELTREE, DISKCOPY).
    - *Specific Commands*: `DEL` deletes files; `DIR` displays a list of files and sub-directories; `DISKCOPY` makes an exact copy of a floppy disk; `LABEL` creates, changes, or removes the volume label of a disk.

2. Basics of Microsoft Office: MS Word

- **Default File & Extension**: Microsoft Word is a word processing program. When started, the default blank document is named **Document1.docx**. The default file extension for Word 2007 and onwards is `.docx`.
- **Window Components**:
    - *Title Bar*: Appears at the very top, showing the application and active file name, containing Minimize, Restore, and Close buttons.
    - *Ruler*: Top and side rulers allow horizontal or vertical alignment formatting.
- **Ribbon Tabs**:
    - *Home*: Contains Clipboard (Cut, Copy, Paste), Font (Size, Color, Bold, Italic, Underline), Paragraph (Bullets, Numbering, Indent), Styles, and Editing (Find, Replace).
    - *Insert*: Pages (Blank Page, Page Break), Tables, Illustrations (Picture, Shapes, SmartArt, Chart), Links (Hyperlink, Bookmark), Header & Footer, Text (TextBox, Date & Time, Object), and Symbols.
    - *Page Layout*: Themes, Page Setup, Page Background, Paragraph, and Arrange.
    - *Review*: Proofing (Spelling & Grammar, Thesaurus, Translate), Comments, Tracking, and Protection. Spelling Check is located here.
    - *View*: Document Views (Print Layout, Full Screen Reading), Zoom, Window, and Macros.
- **Key Document Properties**:
    - *Indentation*: Denotes the distance between text boundaries and page margins (Positive, Hanging, and Negative indents).
    - *Page Orientation*: Portrait (tall, vertical orientation) and Landscape (wide, horizontal orientation).
    - *Thesaurus*: A comprehensive built-in dictionary offering synonym options for a word.
    - *Object Linking and Embedding (OLE)*: A program integration technology used to share information between programs through objects (charts, equations, pictures, sound/video clips).
- **MS Word Keyboard Shortcuts**:
    - `Ctrl + N`: Create new document.
    - `Ctrl + O`: Open existing document.
    - `Ctrl + S`: Save active document.
    - `F12`: Open "Save As" dialog box.
    - `Ctrl + P`: Print document.

    - `Ctrl + F2`: Display Print Preview (displays full pages as printed).
    - `F7`: Check spelling.
    - `Ctrl + X`, `Ctrl + C`, `Ctrl + V`: Cut, Copy, and Paste.
    - `Ctrl + Z`, `Ctrl + Y`: Undo and Redo last action.
    - `Ctrl + K`: Insert hyperlink.
    - `Ctrl + Enter`: Force a page break at the cursor position.
    - `Ctrl + 5`: Increase line spacing to 1.5.
    - `Ctrl + E`: Center alignment.

3. Basics of Microsoft Office: MS Excel

- **Workbook vs. Worksheet**: MS Excel is a spreadsheet calculation software. A **workbook** is an Excel document containing one or more sheets; each new workbook contains **three worksheets by default**. A **worksheet** is a grid of vertical columns (identified by letters: A, B... Z, AA...) and horizontal rows (identified by numbers: 1, 2, 3...).
- **Active Cell & Formula Bar**:
    - The **Active Cell** is the cell in which you are currently working, highlighted by a bold boundary (also specified by a cell pointer).
    - The **Formula Bar** allows entering and displaying values or mathematical equations (formulas) of the active cell.
    - All Excel formulas must begin with an **equal sign (=)**.
- **Cell Address & Referencing**:
    - *Cell Address*: Intersection of column letter and row number (e.g., A1).
    - *Absolute Cell Reference*: The dollar sign (`$`) locks cell locations in a formula (e.g., `$A$4` is absolute; `$A4` is a mixed cell reference where only column A is locked).
- **Core Excel Functions**:
    - `SUM`: Adds all values provided as arguments.
    - `AVERAGE`: Calculates the average of values.
    - `COUNT`: Counts the number of cells that contain numbers.
    - `MAX` & `MIN`: Returns the maximum or minimum value in a list.
    - `TODAY()`: Enters the system date only.
    - `LEN`: Calculates the total characters in a cell.
- **Charts**: Pictorial representations of worksheet data.
    - *Area Chart*: Emphasizes the magnitude of change over time.
    - *Column Chart*: Shows data changes over time, categories arranged horizontally and values vertically.
    - *Bar Chart*: Compares individual items; categories organized vertically, values horizontally.
    - *Line Chart*: Shows trends in data at equal intervals over time.
    - *Pie Chart*: Shows the relationship of parts to a whole (only **one data series** can be plotted).
    - *XY (Scatter) Chart*: Plots pairs of values as coordinates to show relationships among numeric data series.
    - *Chart components*: Chart Area, Plot Area, Chart Title, Axis Title, Gridlines, and Legends.
- **MS Excel Keyboard Shortcuts**:
    - `F2`: Edit selected cell.

- `F11`: Create chart.
- `Ctrl + ;`: Enter current date.
- `Ctrl + Shift + ;`: Enter current time.
- `Alt + Shift + F1` or `Shift + F11`: Insert a new worksheet.
- `Ctrl + Spacebar`: Select the entire column.
- `Shift + Spacebar`: Select the entire row.
- `Ctrl + 9`: Hide the entire row.

4. Basics of Microsoft Office: MS PowerPoint

- **Slides**: PowerPoint is presentation graphics software used to design slides to deliver information visually. The default extension of PowerPoint 2007 and onwards is `.pptx`.
- **PowerPoint Views**:
  - *Normal View*: The primary editing view where you write and design slide contents.
  - *Slide Sorter View*: Displays slides as thumbnails, making it easy to sort, organize, and set transition effects.
  - *Notes Page View*: Located below the slide pane, allowing notes to be typed for each slide.
  - *Slide Show View*: Delivers the presentation to the audience in full-screen mode.
  - *Master View*: Stores slide layout, background colors, placeholder sizes, and text positioning for uniform appearance.
- **Key Media & Design Features**:
  - *Zoom Limit*: PowerPoint can zoom to a maximum of **400% only**.
  - *Media Formats*: Image formats (`.gif`, `.bmp`, `.png`, `.jpg`) and sound formats (`.wav`, `.mid`) can be added.
- **MS PowerPoint Keyboard Shortcuts**:
  - `F5`: View the Slide Show.
  - `Esc`: End or exit the Slide Show.
  - `S`: Stop the slide show (press again to restart).
  - `Ctrl + M`: Insert a new slide.
  - `Shift + Click` or `Ctrl + Click`: Select more than one slide.

Part 2: 25 High-Yield Expected Questions & PYQs

Q1. Which program acts as an organized collection of specialized programs that control overall operations and is necessary for computer booting? [IBPS Clerk]

(1) Utility software
(2) System hardware
(3) Operating System
(4) Application software
Answer: (3)
Explanation: An Operating System acts as the core interface program that initializes hardware and coordinates software execution.

Q2. What process checks to ensure the physical components of the computer are operating and connected properly? [SBI PO]

(1) Editing
(2) Saving
(3) Processing
(4) Booting
Answer: (4)
Explanation: Booting is the initialization process of the computer that copies OS files to RAM and tests connected peripherals.

Q3. Restarting a computer system when it is already powered ON is referred to as:

(1) Cold booting
(2) Warm booting
(3) Hard booting
(4) System logging
Answer: (2)
Explanation: Warm booting refers to restarting an already active system without shutting down physical power.

Q4. MS-DOS uses commands typed sequentially to instruct the computer. What type of user interface is this?

(1) Graphical User Interface (GUI)
(2) Character User Interface (CUI) / Command Line Interface (CLI)
(3) Touch-based Interface
(4) Visual Interface
Answer: (2)
Explanation: MS-DOS is text-based and falls under the CUI/CLI category.

Q5. Which of the following MS-DOS commands is classified as an external command because it requires a file to execute?

(1) VOL
(2) VER
(3) COPY
(4) FORMAT
Answer: (4)
Explanation: FORMAT, CHKDSK, and LABEL are external commands, while VOL, VER, and COPY are internal commands.

Q6. In MS-DOS, what command is used to displays a list of files and sub-directories present in your current folder?

(1) DEL
(2) COPY
(3) DIR
(4) LABEL
Answer: (3)

Explanation: The DIR command displays a complete list of files and sub-directories within the designated path.

Q7. In Microsoft Word, the default blank document created is automatically named:

(1) BlankDoc.docx
(2) Document1.docx
(3) WordPage1.doc
(4) DOC1.doc
Answer: (2)
Explanation: When opening MS Word, the blank document template defaults to Document1.docx.

Q8. Spelling Check in Microsoft Word is a core feature that is located under which ribbon tab? [IBPS PO]

(1) Home tab
(2) References tab
(3) Review tab
(4) Page Layout tab
Answer: (3)
Explanation: The Review tab houses proofing tools, including Spelling & Grammar check and the Thesaurus.

Q9. What tool contains a comprehensive dictionary and offers synonym options for highlighted words in a Word document?

(1) Spell Check
(2) Thesaurus
(3) AutoCorrect
(4) Grammar Check
Answer: (2)
Explanation: Thesaurus is used to search for synonyms of a word in a document.

Q10. What program integration technology allows users to share charts, equations, pictures, and audio clips directly between Office programs?

(1) Object Linking and Embedding (OLE)
(2) Hyperlinking
(3) Rich Text Formatting (RTF)
(4) AutoText
Answer: (1)
Explanation: OLE is used to share and integrate complex objects among Microsoft Office suites.

Q11. Which MS Word keyboard shortcut is used to display the full page preview as it will look when printed?

(1) Ctrl + F1
(2) Shift + F2

(3) Ctrl + F2
(4) Alt + F2
Answer: (3)
Explanation: Ctrl+F2 displays the full-page Print Preview layout of a document.

Q12. In Microsoft Word, you can force a page break at the cursor position by using which shortcut? [IBPS PO]

(1) Alt + Enter
(2) Ctrl + Enter
(3) Shift + Page Down
(4) Ctrl + Page Up
Answer: (2)
Explanation: Ctrl+Enter immediately inserts a page break, moving the cursor to the next page.

Q13. How many worksheets are contained by default when you create a new workbook in Microsoft Excel?

(1) One sheet
(2) Two sheets
(3) Three sheets
(4) Five sheets
Answer: (3)
Explanation: By default, MS Excel populates three worksheets (Sheet1, Sheet2, Sheet3) inside each new workbook.

Q14. In Excel, a cell having a bold boundary that designates where you are currently typing is called a(n):

(1) Range cell
(2) Formula cell
(3) Active cell
(4) Relative cell
Answer: (3)
Explanation: The active cell is designated by a bold outline indicating it is selected for active typing.

Q15. Every formula in Microsoft Excel must begin with which character? [RBI Grade B]

(1) Plus sign (+)
(2) Dollar sign ($)
(3) Slash character (/)
(4) Equal sign (=)
Answer: (4)
Explanation: A formula must begin with an equal sign (=) for Excel to compute the dynamically generated results.

Q16. Which of the following formulas in Excel is NOT valid? [IBPS Clerk]

(1) = A1 + A2
(2) = 1 + A2
(3) = 1A + 2
(4) = A2 + 1
Answer: (3)
Explanation: Cells are addressed by their column letter first and row number second (e.g., A1, A2). "1A" is not a valid cell reference.

Q17. Which function in Excel calculates the count of cells that contain numeric values?

(1) SUM
(2) COUNT
(3) AVERAGE
(4) MAX
Answer: (2)
Explanation: The COUNT function evaluates the provided arguments and counts cells that contain numbers.

Q18. Cell address $A4 in an Excel formula is classified as what type of cell reference?

(1) Mixed cell reference
(2) Absolute cell reference
(3) Relative cell reference
(4) Passive cell reference
Answer: (1)
Explanation: A mixed cell reference has only one component (row or column) locked using a dollar sign ($).

Q19. Which of the following charts is designed to display proportional size of parts to a whole, restricting you to only one data series?

(1) XY Scatter Chart
(2) Bar Chart
(3) Pie Chart
(4) Column Chart
Answer: (3)
Explanation: A pie chart shows parts of a whole and is limited to plotting only one data series.

Q20. In MS Excel, the shortcut key Ctrl + Spacebar is designed to accomplish which of the following?

(1) Hide the active row
(2) Select the entire column
(3) Select the entire row
(4) Delete the highlighted cell
Answer: (2)
Explanation: Ctrl+Spacebar selects the entire column, whereas Shift+Spacebar selects the entire row.

Q21. Which MS Excel shortcut inserts a new worksheet in your workbook?

(1) Ctrl + W
(2) Alt + Shift + F1 / Shift + F11
(3) Ctrl + N
(4) Ctrl + IW
Answer: (2)
Explanation: Alt+Shift+F1 or Shift+F11 is the standardized shortcut to insert a new blank worksheet.

Q22. Which PowerPoint view displays each slide of the presentation as a thumbnail, allowing easy rearrangement? [IBPS Clerk]

(1) Notes Page view
(2) Master view
(3) Slide Show view
(4) Slide Sorter view
Answer: (4)
Explanation: Slide Sorter view displays slide thumbnails and is utilized for organizing slides and transitions.

Q23. What is the maximum zoom percentage limit allowed when working in Microsoft PowerPoint? [IBPS Clerk]

(1) 100%
(2) 200%
(3) 400%
(4) 500%
Answer: (3)
Explanation: Microsoft PowerPoint restricts its maximum magnification zoom to 400%.

Q24. In MS PowerPoint, which shortcut is used to start the slide show from the very first slide?

(1) F1
(2) F2
(3) F5
(4) F10
Answer: (3)
Explanation: Pressing F5 launches the slide show view starting from the first slide.

Q25. Which keyboard key is used to end a running PowerPoint slide show and return to editing mode?

(1) Spacebar
(2) Enter key
(3) Esc key
(4) Backspace key
Answer: (3)
Explanation: The Esc key immediately terminates full-screen slide show view.

📂 I can compile a high-yield theory and practice set for **Chapter 5: Working with Internet and e-mails** (covering web browsing, downloading/uploading, managing e-mail accounts, and e-Banking) when you are ready to continue!

Chapter 5: Working with Internet and e mails

This chapter covers the essentials of digital communication and connectivity, detailing how users search and browse the web, transfer files, manage email communication, and conduct online financial transactions.

Part 1: Core Theory & Source Content

1. Web Browsing & Searching

- **The Internet**: A worldwide network of networked computers that are able to exchange information with each other. It has emerged as a vital and highly efficient means of communication. A host on the Internet is located by other hosts using its unique **IP (Internet Protocol) address**.
- **Web Browser**: A software program used to view Web pages and jump from one page to another. Billions of these digital documents (Web pages) make up the **World Wide Web (WWW)**.
- **Uniform Resource Locator (URL)**: The web address of a document or page on the World Wide Web.
  - *Absolute URL*: Contains the complete address of a file on the Internet.
  - *URL Structure*: A valid URL must contain a protocol identifier (such as `http`), the letters `www`, and the unique registered domain name.
- **Domain Extensions**: The last three letters of a domain name describe the type of organization. Common extensions include:
  - `.edu`: Used for educational institutions.
  - `.com`: Used by commercial or profit businesses.
- **Search Engines**: Specialized software programs that assist users in locating information on the Web. A search engine consists of three key components:
  1. *Spiders or Web Crawlers* (which scour the Web).
  2. *Indexing Software* (which catalogs information).
  3. *Search Algorithms* (which retrieve matching results).
- **Website Navigation**:
  - *Website*: A location on the World Wide Web that is a collection of related Web pages.
  - *Home page*: The very first page that is seen on opening a Website.
  - *Hyperlink*: A reference or connection to data on another Web page that a reader can directly follow or open when clicked or hovered over.
  - *Bookmark*: A stored link to a Web page that allows quick and easy access later.

    - *Cookie*: A small message given to a Web browser by a Web server that stores information about the user's Web activity and is most commonly used to identify return visitors to a website.
    - *Telnet*: A protocol used to establish a **remote login** session on another computer.

2. Downloading & Uploading

- **Data Transfer**: The process of moving files from a remote server on the Internet to a local computer is **downloading**, while sending data from a local machine to a remote system is **uploading**.
- **Podcast**: A program, either talk or numeric, that is made available in digital format for automatic download over the Internet.
- **Web Publishing**: Documents converted to **HTML** (HyperText Markup Language) can be published to the Web.
- **Point-to-Point Protocol (PPP)**: A dial account that puts a computer directly on the Internet using a modem, transmitting data at **9600 bits per second**.
- **Internet Connection Types**:
    - *DSL (Digital Subscriber Line)*: A common type of high-speed **Broadband** connection.
    - *Wi-Fi*: Networks that can be used for public wireless Internet access at wireless hotspots (e.g., restaurants, coffee shops).

3. Managing an E-mail Account

- **E-mail (Electronic-mail)**: A widely used means of personal communication over the Internet.
- **E-mail Client vs. Webmail**:
    - *E-mail Client*: Software like **Microsoft Outlook** which acts as a personal information manager and E-mail client as part of the Microsoft Office suite.
    - *Webmail Interface*: An interface that allows users to access their E-mail accounts from anywhere over the Web.
- **E-mail Address Structure**: Typically consists of a user ID followed by the `@` **(at sign)** and the domain name of the mail provider. Space characters are not allowed.
- **E-mail Folders**:
    - *Drafts*: Retains copies of messages that a user has started but is not yet ready to send.
- **Formatting & Customization**:
    - *Rich Text / Rich Format*: Used if a sender of an E-mail wants to format their message (using bold, italics, colors, etc.). Plain text format does not support these options.
    - *Address Book*: A tool used to store email contacts, allowing users to avoid memorizing complex E-mail addresses.

4. e-Banking

- **e-Banking / Internet Banking**: Conducting financial transactions, checking balances, and paying bills digitally over a secure Internet connection. It is done over the Internet using computers, laptops, or mobile devices.

- **E-commerce (Electronic Commerce)**: Sharing business information, maintaining business relationships, and conducting the process of buying, selling, and trading goods/services over the Internet.
    - *EDI (Electronic Data Interchange)*: The electronic transfer of structured business transactions between a sender and receiver computer.
- **M-commerce (Mobile Commerce)**: Buying and selling goods/services through wireless handheld devices. Mobile commerce was launched in **1997** by Kevin Duffey.

Part 2: 25 High-Yield Expected Questions & PYQs

These multiple-choice questions reflect the exact testing patterns seen in APPSC, AP-TRANSCO, and other public service board exams.

Q1. The Internet is best defined as a:

(1) local area network for a business
(2) software program used to browse documents
(3) worldwide network of networked computers that exchange information
(4) secure operating system
Answer: (3)
Explanation: The Internet is a global network linking smaller networks worldwide, enabling information exchange.

Q2. How does a host computer on the Internet locate another host computer? [RBI Grade B]

(1) By its postal address
(2) By its physical dimensions
(3) By its IP address
(4) By its motherboard serial number
Answer: (3)
Explanation: Hosts on the Internet are identified and located using their unique IP (Internet Protocol) addresses.

Q3. Which of the following is a software program specifically designed to view Web pages? [SBI Clerk]

(1) Device driver
(2) Web browser
(3) Search engine
(4) System utility
Answer: (2)
Explanation: A browser is a software utility that allows users to access, retrieve, and view Web pages.

Q4. What is the literal meaning of the acronym "URL" in web technology? [IBPS Clerk]

(1) Uniform Resource Locator
(2) Unique Read Locator
(3) Unicode Research Location
(4) United Resource Locator
Answer: (1)
Explanation: URL stands for Uniform Resource Locator, which identifies a resource on the Web.

Q5. Which of the following must be contained within a standard URL? [IBPS PO]

(1) The letters WWW only
(2) A protocol identifier and the domain name
(3) A protocol identifier, WWW, and the unique registered domain name
(4) An email address with the `@` symbol
Answer: (3)
Explanation: A standard complete URL contains a protocol identifier (such as HTTP), the WWW subdomain, and the domain name.

Q6. Which type of URL contains the complete address of a file on the Internet? [SSC CGL]

(1) Relative URL
(2) Absolute URL
(3) Static URL
(4) Domain URL
Answer: (2)
Explanation: An absolute URL contains the entire address, including protocol, server domain, and directory path.

Q7. An educational institution would generally have which of the following in its domain name? [IBPS Clerk]

(1) .com
(2) .org
(3) .edu
(4) .net
Answer: (3)
Explanation: The extension `.edu` designates an educational institution's web domain.

Q8. Profit-making businesses and commercial enterprises typically use which domain extension? [SBI Clerk]

(1) .org
(2) .edu
(3) .mil
(4) .com
Answer: (4)
Explanation: The `.com` domain extension is generically used by commercial and profit-seeking organizations.

Q9. Which of the following is NOT one of the three primary components that form a search engine?

(1) Spiders or web crawlers
(2) Indexing software
(3) Search algorithm
(4) Word processing software
Answer: (4)
Explanation: Spiders, indexing software, and search algorithms comprise search engines; word processors are separate applications.

Q10. A stored link to a Web page that a user saves for quick and easy access later is called a(n): [RBI Grade B]

(1) Cookie
(2) Hyperlink
(3) Bookmark
(4) Anchor
Answer: (3)
Explanation: Bookmarks are saved URLs that allow users to return to their favorite Web pages instantly.

Q11. What is the name of the very first page that is seen upon opening a Website?

(1) Web page
(2) Home page
(3) Index page
(4) Template page
Answer: (2)
Explanation: The home page is the initial landing page of any standard website.

Q12. What is a "cookie" in the context of web browsing? [IBPS Clerk]

(1) A virus designed to delete system memory
(2) A hardware key used to secure internet lines
(3) A small message/file that stores information about the user's Web activity
(4) An internet connection protocol
Answer: (3)
Explanation: Cookies are files created by web servers to store browsing history and identify return visitors to a site.

Q13. Telnet is a protocol used for which of the following actions?

(1) Setting up Wi-Fi networks
(2) Remote login
(3) Checking spelling in email attachments
(4) Sending images via e-banking
Answer: (2)
Explanation: Telnet is a terminal emulation protocol that allows a user to log in remotely to a host computer.

Q14. A program, either talk or numeric, made available in digital format for automatic download over the Internet is called a(n): [IBPS Clerk / RRB PO]

(1) Vodcast
(2) Blog
(3) Podcast
(4) Wiki
Answer: (3)
Explanation: Podcasts are audio or video programs released in digital formats for automatic downloading.

Q15. To be published on the Web, documents are formatted and converted to:

(1) Machine language
(2) DOCX files
(3) HTML
(4) Rich Text Format
Answer: (3)
Explanation: HTML (HyperText Markup Language) is the native language used to format documents for web browsers.

Q16. What is the data transmission speed of a Point-to-Point Protocol (PPP) dial account?

(1) 9600 bits per second
(2) 100 Megabits per second
(3) 1.2 Megabytes per second
(4) 48 bits per second
Answer: (1)
Explanation: Point-to-Point Protocol (PPP) dial-up connections transmit data over modems at 9600 bps.

Q17. Digital Subscriber Line (DSL) is an example of which of the following internet connections?

(1) Dial-up
(2) Broadband
(3) Wireless
(4) Analog
Answer: (2)
Explanation: DSL is a high-speed digital connection classified under Broadband technologies.

Q18. Public internet access at hotspots like restaurants and coffee shops is achieved using:

(1) Wi-Fi networks
(2) DSL lines
(3) Co-axial cabling
(4) Parallel ports

Answer: (1)
Explanation: Wi-Fi networks provide local, high-speed wireless internet access to devices in public hotspots.

Q19. Which application in the Microsoft Office suite is built to serve as an E-mail client?

(1) MS Access
(2) MS Excel
(3) MS PowerPoint
(4) MS Outlook
Answer: (4)
Explanation: Microsoft Outlook is the dedicated E-mail client and personal information manager in MS Office.

Q20. What interface is designed to allow a user to access their e-mail from anywhere in the world?

(1) Plain format
(2) Webmail interface
(3) Message board
(4) Forum
Answer: (2)
Explanation: Webmail interfaces let users access and manage email directly via a web browser from any online machine.

Q21. An E-mail address typically consists of a user ID followed by which symbol?

(1) Hash symbol (`#`)
(2) At sign (`@`)
(3) Ampersand (`&`)
(4) Dollar sign (`$`)
Answer: (2)
Explanation: E-mail addresses are structured as: `UserID @ DomainName`.

Q22. While composing an email, where are messages retained if they are started but not yet ready to be sent?

(1) Inbox folder
(2) Outbox folder
(3) Sent Items folder
(4) Drafts folder
Answer: (3)
Explanation: The drafts folder holds half-finished or unsent outgoing email messages.

Q23. If an E-mail sender wants to utilize formatting like bold or italic text, which format must they use? [IBPS Clerk]

(1) Plain text
(2) Rich text / Rich format

(3) ASCII format
(4) Read-only format
Answer: (2)
Explanation: Rich text/formatting supports styling elements like bolding, italics, and custom fonts, which plain text does not.

Q24. What is the electronic transfer of structured business transactions between computer systems called?

(1) E-commerce
(2) Electronic Data Interchange (EDI)
(3) Mobile commerce
(4) Internet telephony
Answer: (2)
Explanation: EDI is the electronic exchange of standardized business data between computer applications.

Q25. Who launched Mobile Commerce (M-commerce) in the year 1997?

(1) Kevin Duffey
(2) Ted Hoff
(3) Douglas Engelbart
(4) John Von Neumann
Answer: (1)
Explanation: Mobile commerce (M-commerce) was introduced in 1997 by Kevin Duffey.

📂 I can compile a high-yield theory and practice set for **Chapter 6: Basics of networking and cyber security** (covering network devices, protocols, safety threats like hacking/viruses, and firewalls) when you are ready to continue!

Chapter 6: Basics of networking and cyber security

This final chapter covers how computer networks communicate, the physical and logical architectures that connect them, the protocols that govern data transmission, the cyber threats that target systems, and the preventive measures used to protect information.

Part 1: Core Theory & Source Content

1. Networking Basics, Devices, and Protocols

- **Data Communication**: The exchange of data between two devices using some form of transmission media. Data is transferred in the form of signals, which can be **analog** or **digital**.
- **Communication Channels**:

- *Simplex Channel*: Flow of data is always in one direction (e.g., keyboard to computer).
- *Half Duplex Channel*: Data flows in both directions, but only one direction at a time (e.g., Walkie-Talkie).
- *Full Duplex Channel*: Data flows in both directions simultaneously (e.g., mobile phones).

- **Transmission Media**:
  - *Guided (Wired) Media*: Physical pathways like **Twisted Pair/Ethernet cabling** (low-speed, shielded or unshielded), **Co-axial Cable** (used for multi-channel TV networks), and **Fibre Optic Cable** (transmits data as light waves at exceptionally high speeds, offers high bandwidth, and is immune to electromagnetic interference).
  - *Unguided (Wireless) Media*: **Radio waves** (omnidirectional), **Microwaves** (unidirectional high-frequency electromagnetic waves), **Infrared waves** (used for short-range communication like TV remotes), and **Satellite Communication** (long-distance, high-speed transmission).
- **Bandwidth & Throughput**:
  - *Bandwidth*: The maximum amount of information a communication medium can transfer in a given amount of time, measured in Hertz (Hz).
  - *Throughput*: The actual amount of data transmitted, measured in bits per second (bps). **Gigabits per second (Gbps)** is the fastest standard transmission speed unit.
- **Networking Devices**:
  - *Repeater*: Operates at the physical layer to amplify signals and restore their strength over long distances.
  - *Hub*: A multi-port device that acts as a centralized node, broadcasting incoming data packets to all of its ports. It is typically associated with a **star topology**.
  - *Switch*: A small hardware device that joins multiple computers together within one LAN. It reduces traffic by establishing a temporary connection and directing packets only to the specified recipient. It is often referred to as a "multiport bridge".
  - *Router*: A hardware device designed to analyze, move, convert, direct, or drop incoming packets across different networks.
  - *Gateway*: An interconnecting device that joins two different network protocols together; it acts as a protocol converter.
  - *Bridge*: Serves a similar function to switches, filtering data traffic at a network boundary.
- **Network Topologies** (Geometric arrangement of devices):
  - *Bus Topology*: All components are connected directly to a single main cable.
  - *Star Topology*: Peripheral nodes are connected to a central node/hub.
  - *Ring (Circular) Topology*: High-performance network where data is transmitted sequentially in the form of a **Token**. Each node is connected to exactly two other nodes.
  - *Mesh Topology*: Every node has a dedicated point-to-point link to every other node. It offers the **highest reliability**.
  - *Tree Topology*: Nodes are arranged hierarchically like an inverted tree, branching out from a root server.
- **OSI (Open System Interconnection) Model**: A seven-layer standard reference model for network communication:

1. *Physical Layer (Lowest)*: Handles hardware connections, voltages, data rates, and converts bits to electrical signals. Repeaters operate here.
2. *Data Link Layer*: Error detection/correction, assembles messages into frames. Switches operate here.
3. *Network Layer*: Routing of signals, splits outgoing messages into packets. Routers operate here.
4. *Transport Layer*: Segmenting data, ensuring accurate transmission, and packet filtering.
5. *Session Layer*: Establishes, maintains, and synchronizes conversations.
6. *Presentation Layer*: Acts as a translating layer; handles data formatting, compression, encryption, and decryption.
7. *Application Layer (Highest)*: User interface for retransferring files, login, and password checking. Packet filtering firewalls operate here.

- **Ethernet**: A widely used LAN technology utilizing bus topology. Standard Ethernet operates at 10 Mbps and utilizes a unique **48-bit physical address**.

2. Network and Information Security Threats

- **Hacking**: Unauthorized access to a computer system or network.
- **Malicious Software (Malware)**:
  - *Virus*: A program able to harm computer systems. Unlike normal software, it attaches to files and replicates itself to disrupt operations. If a computer keeps rebooting itself spontaneously, it is highly likely that it has been infected by a virus.
  - *Worms*: Self-replicating malicious programs that spread rapidly across networks.
  - *Trojan (Trojan Horses)*: Malicious applications that masquerade as harmless or user-friendly programs to trick users into executing them.
  - *Bomb*: A specialized type of virus designed to activate its destructive payload only on a specific date or time.
  - *Keylogger*: A software program designed to secretly record (log) every keystroke made on a target machine.
- **Authentication & Password Security**:
  - *Authentication*: The process of verifying a user's identity (such as verifying a username and password) before granting network access.
  - *Password strength*: Safe practices demand complex passwords (e.g., using a combination of letters, numbers, and special characters, like "Timepass_09") rather than easily guessed personal details.

3. Preventive Measures

- **Firewalls**: A security barrier used to prevent unauthorized access. A packet-filtering firewall inspects incoming data packets and operates at the application layer of the OSI model.
- **Antivirus Software**: Programs designed to detect, neutralize, and eliminate computer viruses and malware. Popular antivirus suites include: *Avast, AVG, K7, Kaspersky, Trend Micro, Quick Heal, Symantec, Norton, and McAfee*.
- **Cryptography, Digital Signatures, and Certificates**:
  - *Encryption & Decryption*: Function of the Presentation Layer that encodes data into secret codes to prevent hackers from reading it.

- *Digital Certificate*: An attachment to an electronic message used for security to prove identity in electronic transactions.
- *Digital Signature*: An electronic form of signature used to authenticate the identity of the sender and ensure that the original content of the message has not been altered during transit.

Part 2: 25 High-Yield Expected Questions & PYQs

Q1. Which of the following is defined as the transmission of data or information between two or more computers over communication links?

(1) System Administration
(2) Data communication
(3) Software Processing
(4) Hardware Interfacing
Answer: (2)
Explanation: Data communication involves the exchange of data between separate devices over physical or wireless links.

Q2. In a simplex communication channel, the flow of data is strictly restricted in which manner?

(1) Always in both directions simultaneously
(2) In both directions, but only one direction at a time
(3) Always in one direction
(4) None of the above
Answer: (3)
Explanation: Simplex channels are strictly unidirectional; data can only travel from sender to receiver (e.g., keyboard to CPU).

Q3. Communication between a personal computer and a keyboard involves which type of transmission? [IBPS Clerk Mains]

(1) Automatic
(2) Half-duplex
(3) Full-duplex
(4) Simplex
Answer: (4)
Explanation: Because the keyboard can only send inputs to the computer and cannot receive data back, it represents simplex transmission.

Q4. A mobile phone or wireless handset is an example of which type of communication channel?

(1) Simplex channel
(2) Half-duplex channel
(3) Full-duplex channel
(4) Dial-up channel

Answer: (3)
Explanation: Mobile phones allow both users to talk and listen simultaneously, representing full-duplex communication.

Q5. Which of the following transmission media carries communication signals as light waves and is highly resistant to electromagnetic interference? [IBPS Clerk Mains]

(1) Twisted pair cable
(2) Co-axial cable
(3) Optic fibre cable
(4) Flat ribbon cable
Answer: (3)
Explanation: Fibre optic cables utilize glass or plastic strands to transmit data as pulses of light, giving them massive speed, bandwidth, and resistance to interference.

Q6. What does the term "Bandwidth" refer to in data communication systems? [RBI Grade B]

(1) The total physical cost of cables required to implement a local network
(2) The amount of information a communication medium can transfer in a given amount of time
(3) The physical thickness of co-axial copper wires
(4) The number of active users connected to a web server
Answer: (2)
Explanation: Bandwidth defines the maximum carrying capacity of a communication link, measured in Hertz (Hz).

Q7. Which of the following speed units represents the fastest rate of data transmission? [SBI Clerk]

(1) kbps
(2) mbps
(3) gbps
(4) bps
Answer: (3)
Explanation: Gbps (Gigabits per second) represents a billion bits per second, which is faster than bps, kbps, or mbps.

Q8. What do we call a physical or logical arrangement where two or more computers are connected together by cables or wireless signals to share resources and data? [SBI Clerk]

(1) Operating System
(2) Network
(3) Hyperlink
(4) Gateway
Answer: (2)
Explanation: A network connects multiple nodes/computers together to exchange data and share peripheral hardware.

Q9. A protocol in computer networking is best defined as a(n): [SSC CGL]

(1) Physical cable connection
(2) Device driver for modems
(3) Agreement or set of rules governing how communication is to proceed
(4) Operating system user interface
Answer: (3)
Explanation: Protocols are the standardized sets of rules that define how data is formatted, sent, received, and processed across a network.

Q10. Which networking device operates at the Physical Layer of the OSI model to amplify and restore the original strength of signals?

(1) Router
(2) Repeater
(3) Gateway
(4) Switch
Answer: (3)
Explanation: A repeater receives weak signals over long distances and regenerates them to prevent attenuation.

Q11. Which hardware device is designed to join multiple computers together within a single LAN and forwards packets specifically to the destination port?

(1) Repeater
(2) Gateway
(3) Hub
(4) Switch
Answer: (4)
Explanation: A switch acts as an intelligent multiport bridge, reading MAC addresses to forward data only to the appropriate recipient port.

Q12. What device is required to link your computer with other computers and digital services through analog telephone lines? [SBI Clerk]

(1) Multiplexer
(2) Modem
(3) Gateway
(4) Repeater
Answer: (2)
Explanation: A modem (Modulator-Demodulator) converts digital computer signals into analog telephone signals and vice-versa.

Q13. An interconnecting hardware device that joins two entirely different network protocols together is called a:

(1) Bridge
(2) Switch
(3) Gateway
(4) Hub

Answer: (3)
Explanation: Gateways act as protocol converters, accepting formatted packets from one protocol and translating them for another.

Q14. The geometric or physical arrangement of various devices and cabling on a network is called its:

(1) Protocol
(2) Topology
(3) Media
(4) System Bus
Answer: (2)
Explanation: Topology is the structural map of how nodes are wired or linked together physically and logically.

Q15. In which network topology are all individual components connected directly to a single, continuous main cable?

(1) Star topology
(2) Ring topology
(3) Bus topology
(4) Tree topology
Answer: (3)
Explanation: Bus topology connects all devices to a single common cable (often called the bus or "Ether").

Q16. Which network topology connects all peripheral nodes to a central node or hub? [SBI Clerk]

(1) Ring topology
(2) Mesh topology
(3) Bus topology
(4) Star topology
Answer: (4)
Explanation: In a star network, all communication passes through a central device like a hub or switch.

Q17. In a ring topology, what is the name of the small message or permission packet that a computer must possess to transmit data?

(1) Header
(2) Token
(3) Frame
(4) Packet
Answer: (2)
Explanation: Ring networks use token-passing protocols; only the node holding the "Token" can write data onto the ring.

Q18. Which network topology offers the highest reliability because every node has a dedicated point-to-point link to every other node? [IBPS RRB PO Mains]

(1) Mesh topology
(2) Tree topology
(3) Ring topology
(4) Star topology
Answer: (1)
Explanation: Mesh topology is completely interconnected, meaning if one link fails, traffic can immediately be rerouted through alternative paths.

Q19. How many physical bits constitute a standard hardware Ethernet address? [SBI Clerk]

(1) 16 bits
(2) 32 bits
(3) 48 bits
(4) 64 bits
Answer: (3)
Explanation: An Ethernet (MAC) address is a unique 48-bit hardware identifier hardcoded into the network interface.

Q20. What is the name of the fourth layer of the OSI standard reference model? [SBI PO]

(1) Physical Layer
(2) Session Layer
(3) Transport Layer
(4) Network Layer
Answer: (3)
Explanation: Counted from the bottom (Physical=1, Data Link=2, Network=3), the 4th layer is the Transport Layer.

Q21. In the OSI model, which layer is responsible for signal routing, packet handling, and IP addressing? [IBPS Clerk]

(1) Data Link Layer
(2) Transport Layer
(3) Network Layer
(4) Session Layer
Answer: (3)
Explanation: The Network Layer handles routing across intermediate networks and manages IP packets.

Q22. What process verifies a user's identity (such as checking their username and password) before they can access a secure system?

(1) Logic check
(2) Authorization
(3) Authentication
(4) Encryption
Answer: (3)

Explanation: Authentication is any secure protocol that validates a user's credentials before allowing system entry.

Q23. Which type of virus is uniquely designed to activate its destructive payload only when a specific date or time is reached?

(1) Trojan horse
(2) Worm
(3) Keylogger
(4) Bomb
Answer: (4)
Explanation: A "Bomb" (or time bomb) is malware programmed to lie dormant until triggered by a specific calendar date or clock threshold.

Q24. An electronic security attachment used to authenticate a sender's identity and guarantee that the original message contents have not been modified is a:

(1) Digital Signature
(2) Digital Certificate
(3) Packet Filter
(4) Password Vault
Answer: (1)
Explanation: A digital signature authenticates the sender's identity and preserves data integrity.

Q25. Spontaneous and persistent rebooting of a computer system is a common symptom of which system issue?

(1) Spooler overflow
(2) Processor clock lag
(3) Lack of printer connection
(4) A computer virus
Answer: (4)
Explanation: If a computer keeps rebooting itself without user intervention, it is highly likely infected by a virus.

🎓 We have completed all 6 chapters of your syllabus! Would you like me to generate a **comprehensive final practice exam** combining all chapters to test your overall readiness for those 5 exam marks?